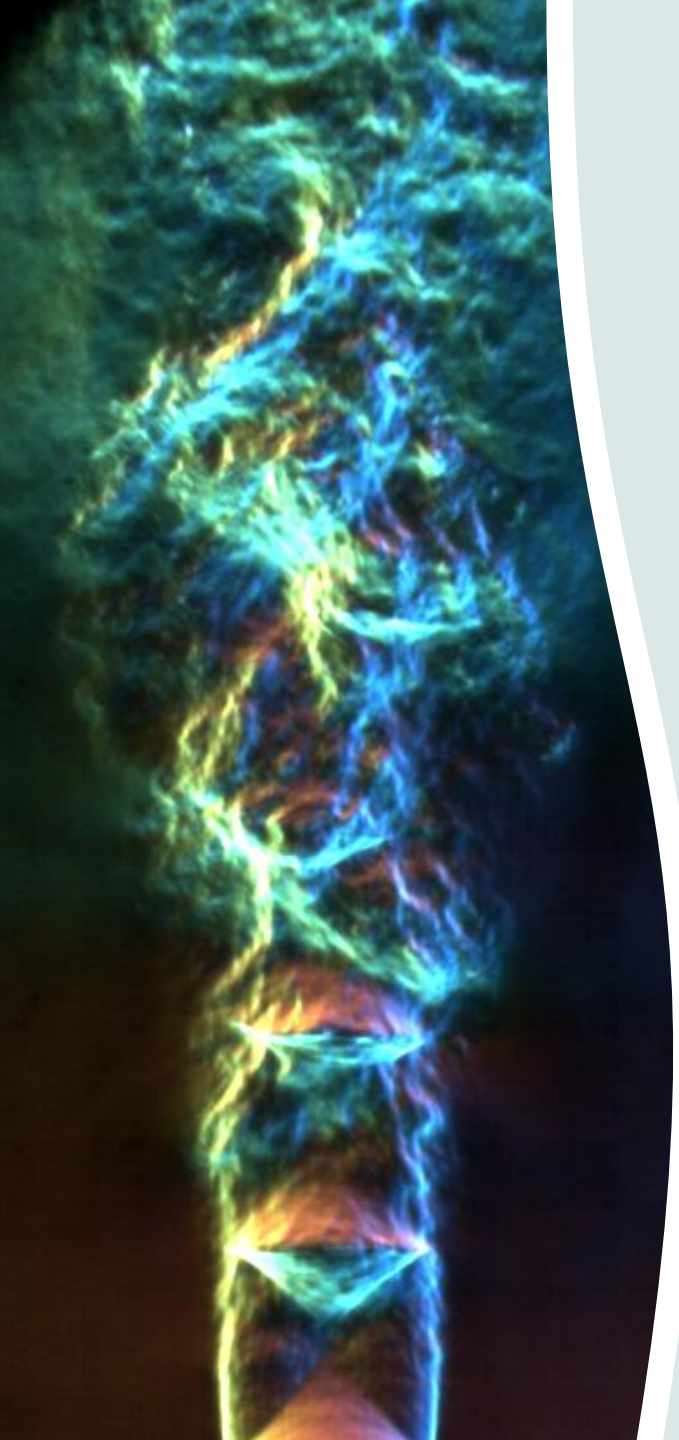




Characterising the Acoustic Properties of Hydrogen Leaks

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Problem

- Hydrogen is rapidly becoming the target of major investment in the long-distance transport industry as well as energy.
- The result of this is large hydrogen farms are needed for storage.
- Hydrogen is well known for its ability to leak through most seals, these can be mitigated but are extremely difficult to eliminate and the leaks can start from anywhere in the system.
- Being able to detect these leaks acoustically before they are detected through standard particle leak detectors could reduce the difficulty of managing these large farms and make safety easier.

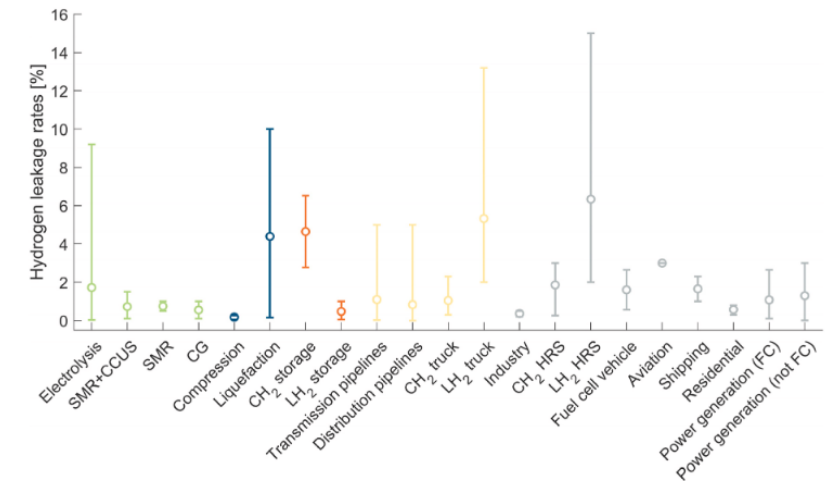
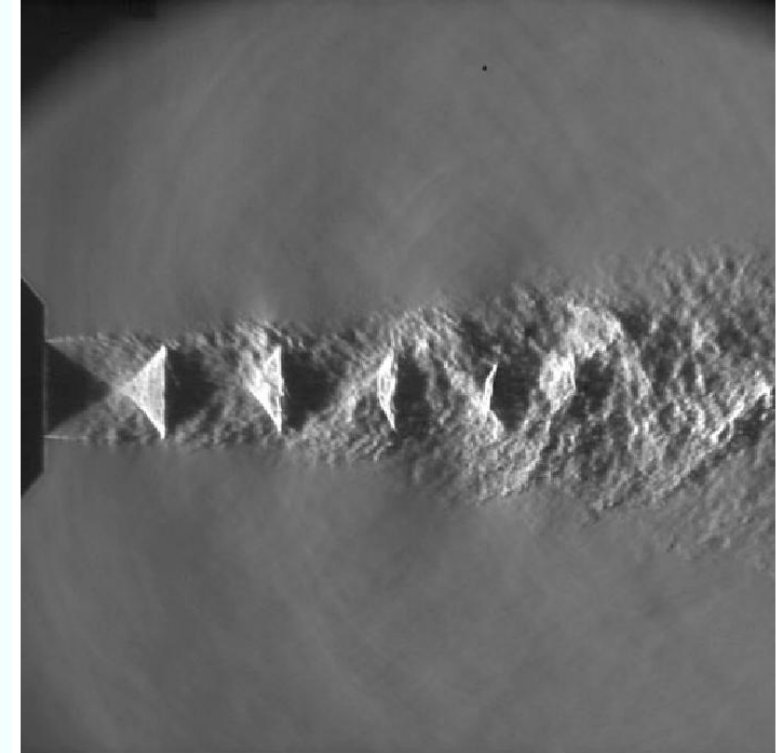


Fig. 2. Hydrogen leakage rates (% mass) of the different stages in the hydrogen supply chain. Average values (depicted as circles) and minimum-maximum variation ranges are shown. The supply chain includes the following stages: production (green color), handling (blue color), storage (orange color), transport (yellow color) and end-uses (grey color). For a complete graphical representation of the hydrogen leakage dataset, the reader is referred to [9]. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Hydrogen leak rates at various stages of the supply chain Davide Trapani et al.

Methodology

- This project aims to detect leaks using acoustic measurements to see if hydrogen has distinct jet screech tones detectable with lab equipment.
- Hydrogen is typically stored as a high pressure compressed gas or sub-cooled cryogenic liquid, either way, above atmospheric pressure.
- It can be assumed all leaks will result in choked flow.
- Acoustic microphones and high speed schlieren will be used for measurement.
- A dedicated hydrogen jet facility will be created and commissioned in the Shock Lab.
- This requires very rigorous safety analysis and review, hydrogen is a dangerous flammable gas.



High speed schlieren imaging.

Problem Analysis

- The challenges of this project remain heavily safety oriented.
- Hydrogen auto-ignition is possible at high pressure ratios.
 - This could be a danger if occurring unexpectedly, this is only expected occur at pressure ratios an order of magnitude higher than what this experiment proposes.
 - This could be beneficial as it will combust any hydrogen before entering the extraction, effectively making it inert.
- High pressures and sounds are inherently dangerous, this system can only be made safe if operated outside of the lab, this will require control hardware and software routed outside the lab.
- There is existing equipment in the lab that this must fit around, this is a given constraint that must be worked around.

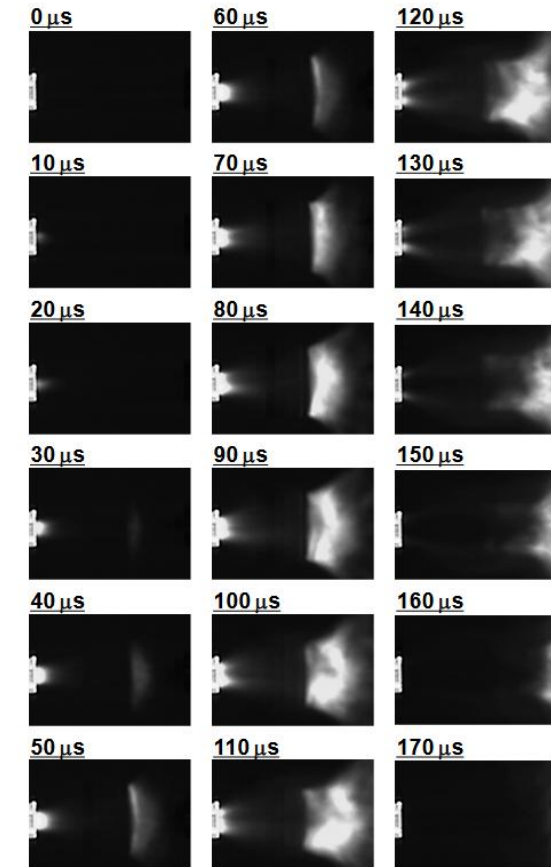


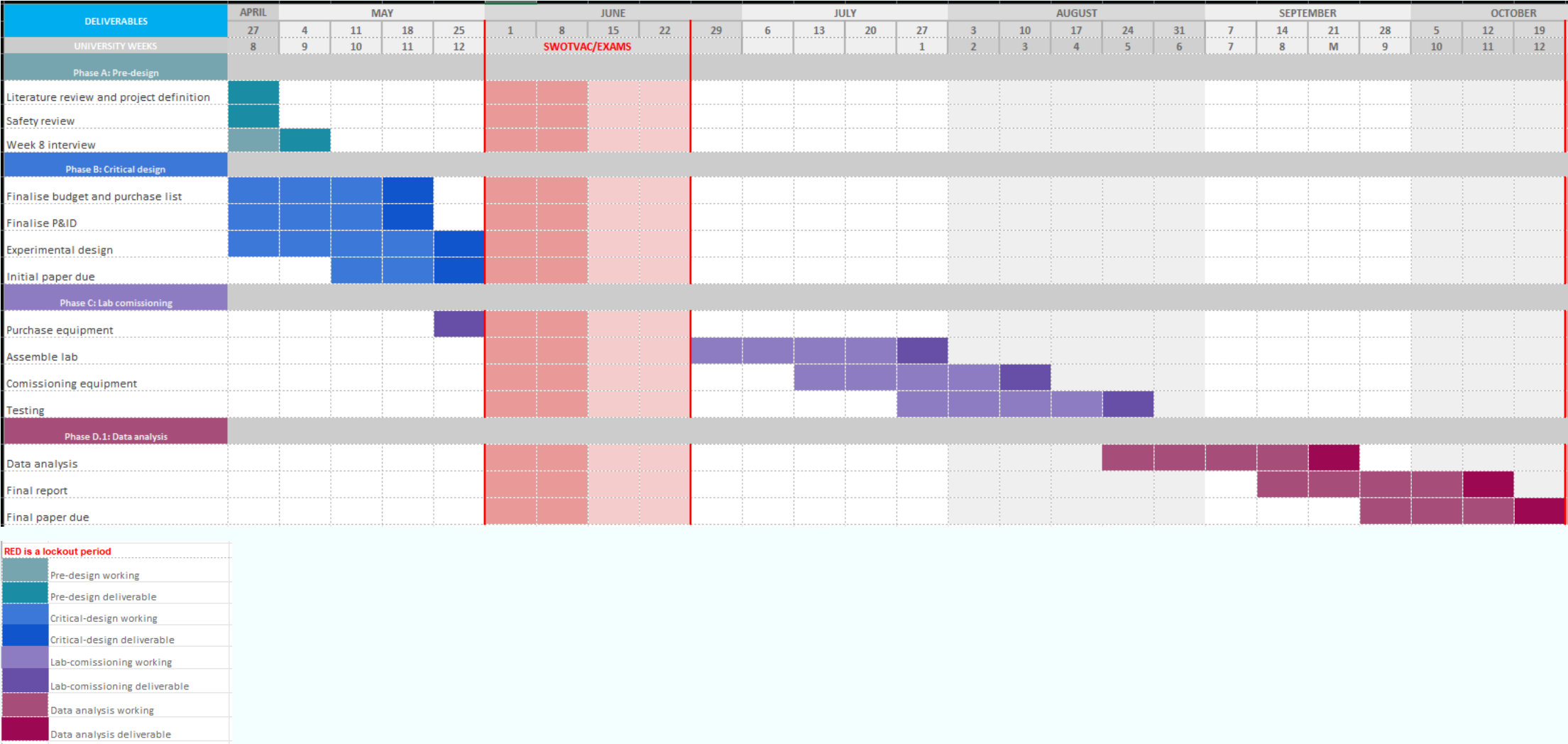
Fig. 5 High speed sequential pictures of the hydrogen jet, flame generation, and its propagation at tube exit of the tube ($L = 650$ mm, $P_b = 5.0$ MPa)

High speed schlieren imaging demonstrating auto-ignition of hydrogen at 50 bar Kitabayashi N et al.

Scope and Timeline

- The aims for this project are:
 - Complete a feasibility study on the safety of creating a hydrogen jet facility at Monash
 - Determine the theoretical feasibility of detecting leaks with acoustics within the bounds of the safety requirements set
 - Design and build a safe hydrogen jet facility
 - Commission the facility and test a hydrogen jet. Perform appropriate data analysis techniques to support or contradict hypothesis

Scope and Timeline

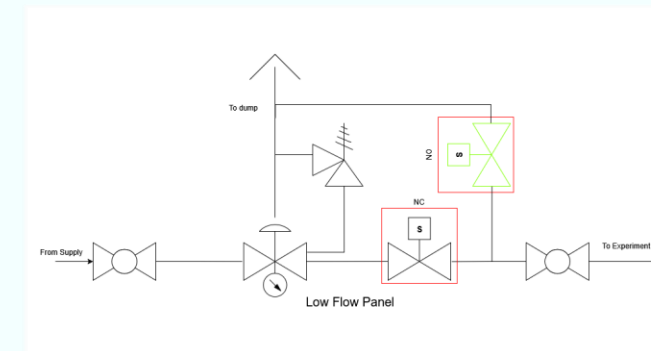
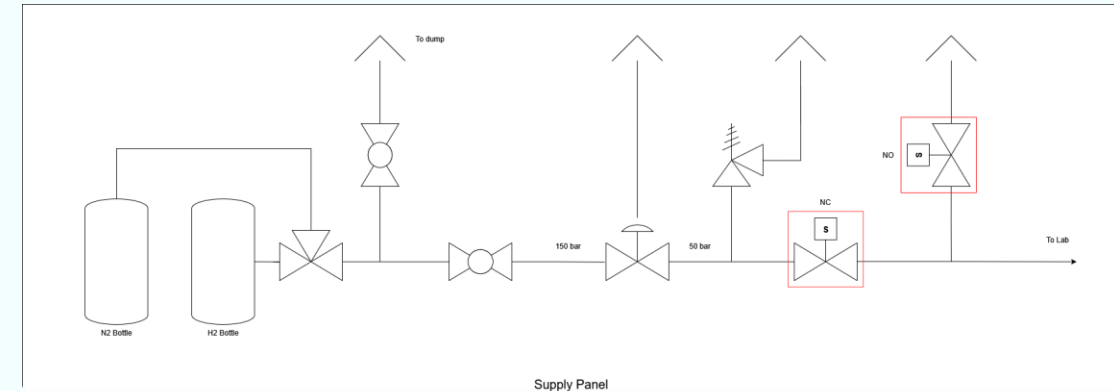


Safety Risks

- The first part of this project has been focused on a safety review, investigating key risk identified:
 1. Hydrogen concentration leaving the jet exceeding the lower explosive limit and combusting around an ignition source.
 2. Hydrogen combusting inside the extraction system
 3. Hydrogen leaking outside of the designated flow path into the room
 4. Being able to immediately shut the system down and safely evacuate all hydrogen from the feed system

Safety Risks

- These issues have been addressed by applying restrictions on nozzle flow rate based the AS/IEC 60079.10.1 standard and the WorkSafe regulations. The key outcomes are:
 - No one to be in the lab during operation
 - Integrated e-stop system to safely stop hydrogen flow and initiate a nitrogen purge
 - Adequate lab extraction
 - Adequate detection in the lab to monitor hydrogen build up



Proposed fluids Piping and Instrumentation Diagram.

Safety Risks

- Per AS/IEC 60079.10.1 standard the environment that has been defined is:
 - “Primary” release – release which can be expected to occur periodically or occasionally during normal operation
 - Type C opening – Openings which are normally closed, infrequently opened and fitting with sealing devices.
 - Dilution high – with expected 19 m/s of extraction velocity we remain a high dilution environment with characteristic flow rate less than 1500 L/s.
 - Zone 1 NE – the result of this is a zone of negligible extent, indicating a non-hazardous area, removing the need for “ex-rated” equipment.
 - Adequate and direct extraction removes risk of background concentration build up, detectors will be used to monitor any stray leaks.

$$Q_c = \frac{W_g}{\rho_g \times LFL}$$

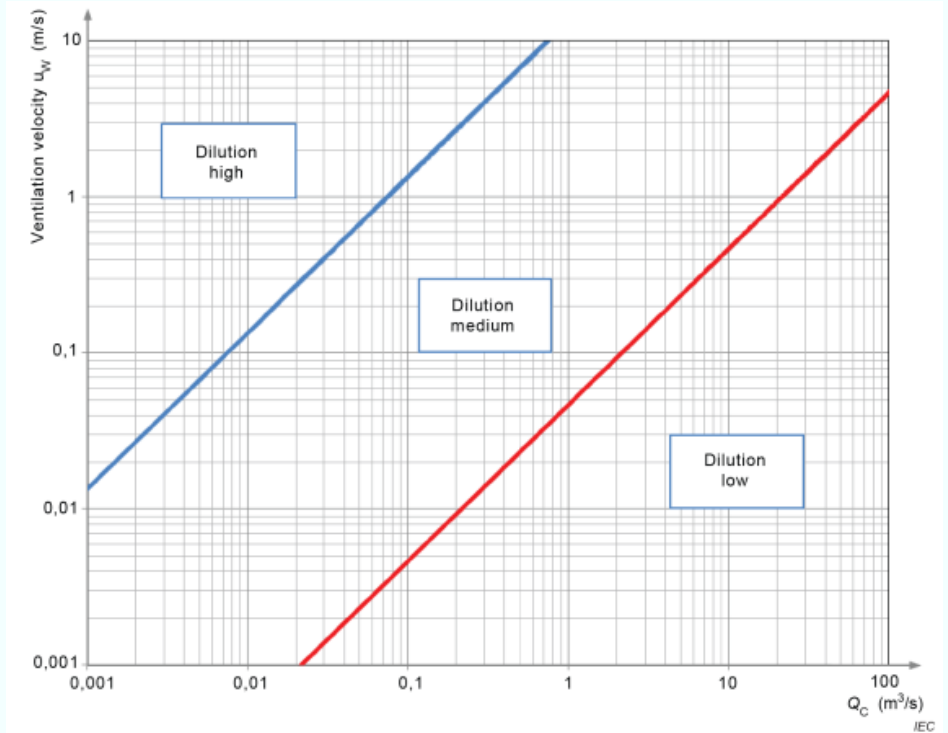


Figure C.1 – Chart for assessing the degree of dilution

AS/IEC 60079.10.1 standard plot of ventilation velocity against characteristic flowrates, with characteristic lines defining dilution grades.

Sizing

- Safety standards require the nozzle to be 3.5 mm in diameter or less to maintain safe extraction across a pressure ratio up to 6.
- Phenomena of interest are expected at Strouhal numbers between 0 and 0.2 for H₂.
- With the equipment available it is expected that a Strouhal number up to 1.46 is possible.

H. Ali et al.

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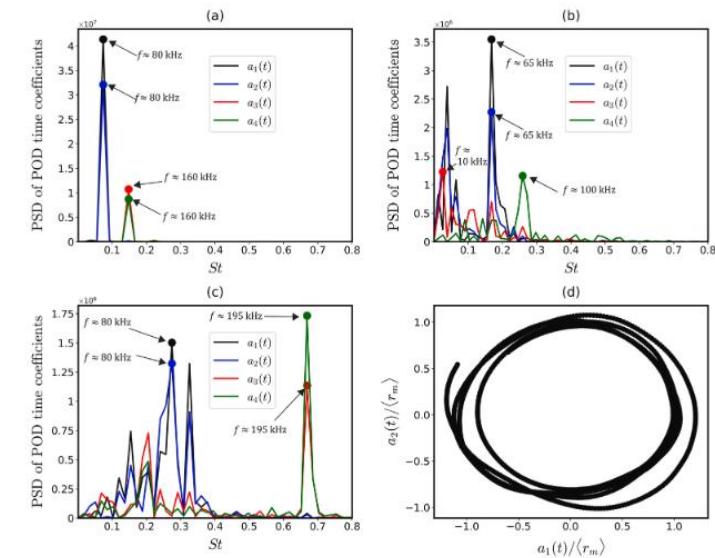
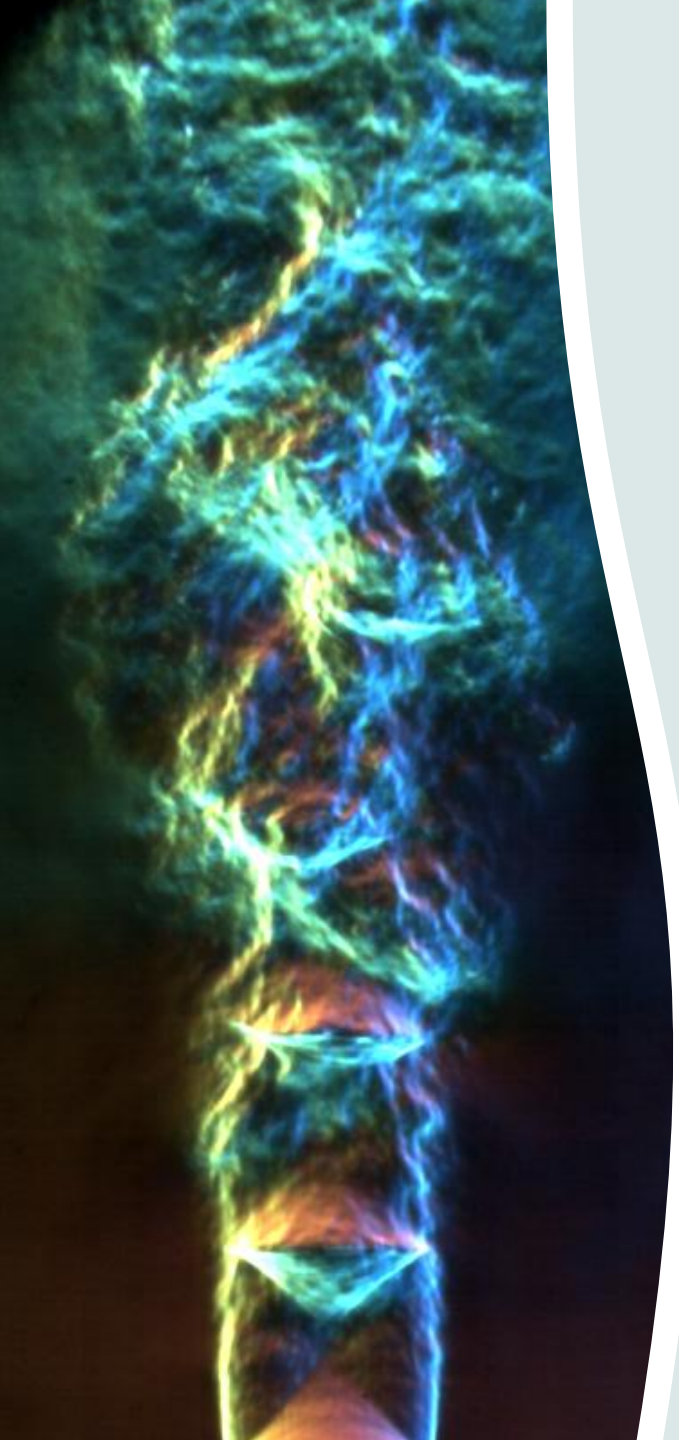


Fig. 6. (a) PSD of time coefficients of first four POD modes for H₂ jet (NPR = 6.5), (b) PSD of time coefficients of first four POD modes for CH₄ jet (NPR = 6.5), (c) PSD of time coefficients of first four POD modes for N₂ jet (NPR = 6.5), and (d) Phase portrait for time coefficients of first two POD modes $a_1(t)$ and $a_2(t)$ for H₂ jet (NPR = 6.5) to reveal pairing where $\langle r_m \rangle = (\sqrt{a_1^2 + a_2^2})$.

PSDs of H₂, CH₄ and N₂ jets Haseeb Ali et al.

FYP A Initial Paper Marking Rubric

Criterion	Pass Standard
Abstract (mandatory)	● A description of the project topic/problem. ● A short discussion of the background/context for this project. ● A short description of the methodology and/or methods. ● Initial and/or anticipated results.
Introduction (mandatory)	● Provide an initial literature review or state of the field. ● Describe the relevance of the project in the context of the previous research. ● Describe the project aims/objectives. ● Includes a research question or problem statement.
Methodology/Methods (Can be under a section of any name)	● Appropriate methods are proposed and justified ● Initial results (Can be under a section of any name) ● Initial and/or anticipated results are presented
Quality of the presentation	● The paper is no longer than 4 pages. ● The academic writing is able to be clearly understood by the marker. ● The paper has a clear structure which makes it straightforward to read. ● The tables and figures (if included) are clear, labelled correctly and referenced in the text. ● All references are cited in the text and all citations in the text are in the reference list. The referencing style is consistent.



Thanks for listening!



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